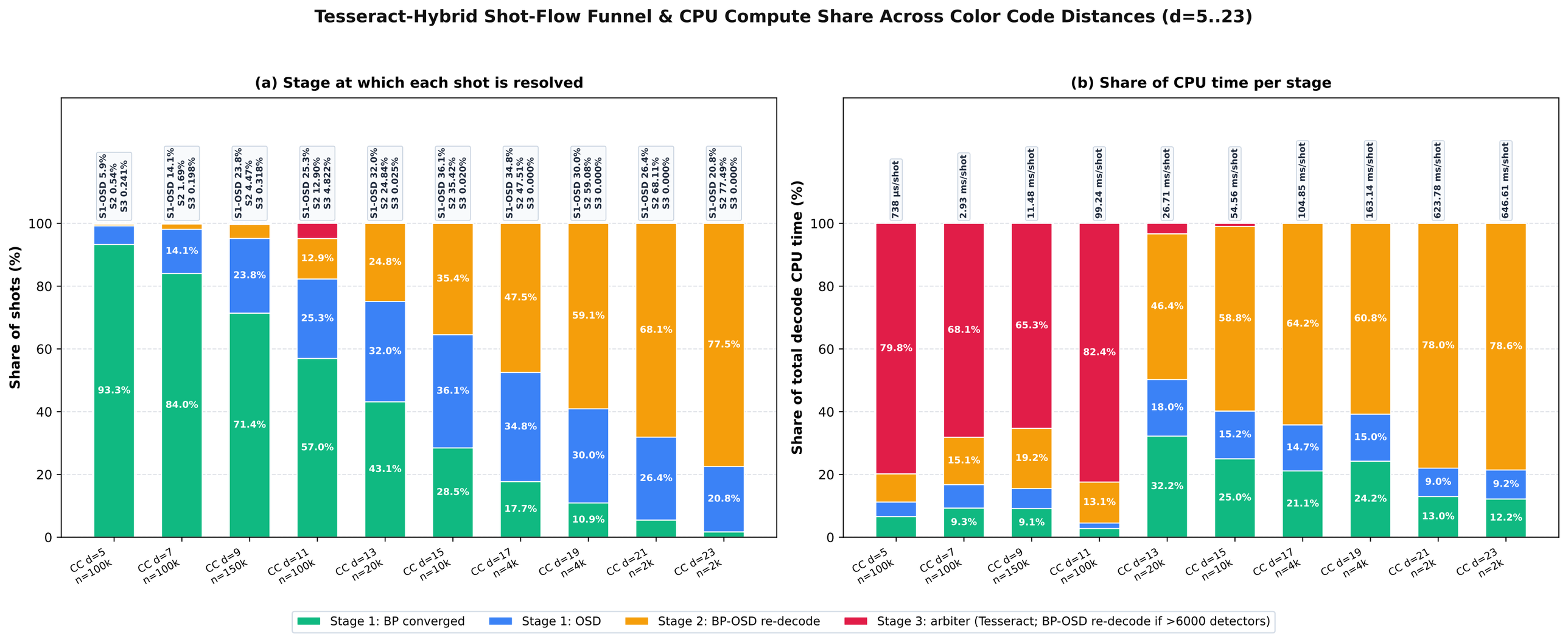
Tesseract-Hybrid Shot-Flow Funnel & CPU Compute Share Across Color Code Distances (d=5..23)
(a) Stage at which each shot is resolved
Share of shots (%)
S1-OSD 5.9%
S2 0.54%
S3 0.241%
S1-OSD 14.1%
S2 1.69%
S3 0.198%
S1-OSD 23.8%
S2 4.47%
S3 0.318%
S1-OSD 25.3%
S2 12.90%
S3 4.822%
S1-OSD 32.0%
S2 24.84%
S3 0.025%
S1-OSD 36.1%
S2 35.42%
S3 0.020%
S1-OSD 34.8%
S2 47.51%
S3 0.000%
S1-OSD 30.0%
S2 59.08%
S3 0.000%
S1-OSD 26.4%
S2 68.11%
S3 0.000%
S1-OSD 20.8%
S2 77.49%
S3 0.000%
93.3%
14.1%
84.0%
23.8%
71.4%
12.9%
25.3%
57.0%
24.8%
32.0%
43.1%
35.4%
36.1%
28.5%
47.5%
34.8%
17.7%
59.1%
30.0%
10.9%
68.1%
26.4%
5.5%
77.5%
20.8%
1.7%
CC d=5 n=100k
CC d=7 n=100k
CC d=9 n=150k
CC d=11 n=100k
CC d=13 n=20k
CC d=15 n=10k
CC d=17 n=4k
CC d=19 n=4k
CC d=21 n=2k
CC d=23 n=2k
(b) Share of CPU time per stage
Share of total decode CPU time (%)
738 µs/shot
2.93 ms/shot
11.48 ms/shot
99.24 ms/shot
26.71 ms/shot
54.56 ms/shot
104.85 ms/shot
163.14 ms/shot
623.78 ms/shot
646.61 ms/shot
79.8%
6.7%
4.4%
9.3%
68.1%
15.1%
9.1%
65.3%
19.2%
9.1%
82.4%
13.1%
3.1%
1.4%
32.2%
18.0%
46.4%
25.0%
15.2%
58.8%
1.2%
21.1%
14.7%
64.2%
24.2%
15.0%
60.8%
13.0%
9.0%
78.0%
12.2%
9.2%
78.6%
Stage 1: BP converged
Stage 1: OSD
Stage 2: BP-OSD re-decode
Stage 3: arbiter (Tesseract; BP-OSD re-decode if >6000 detectors)